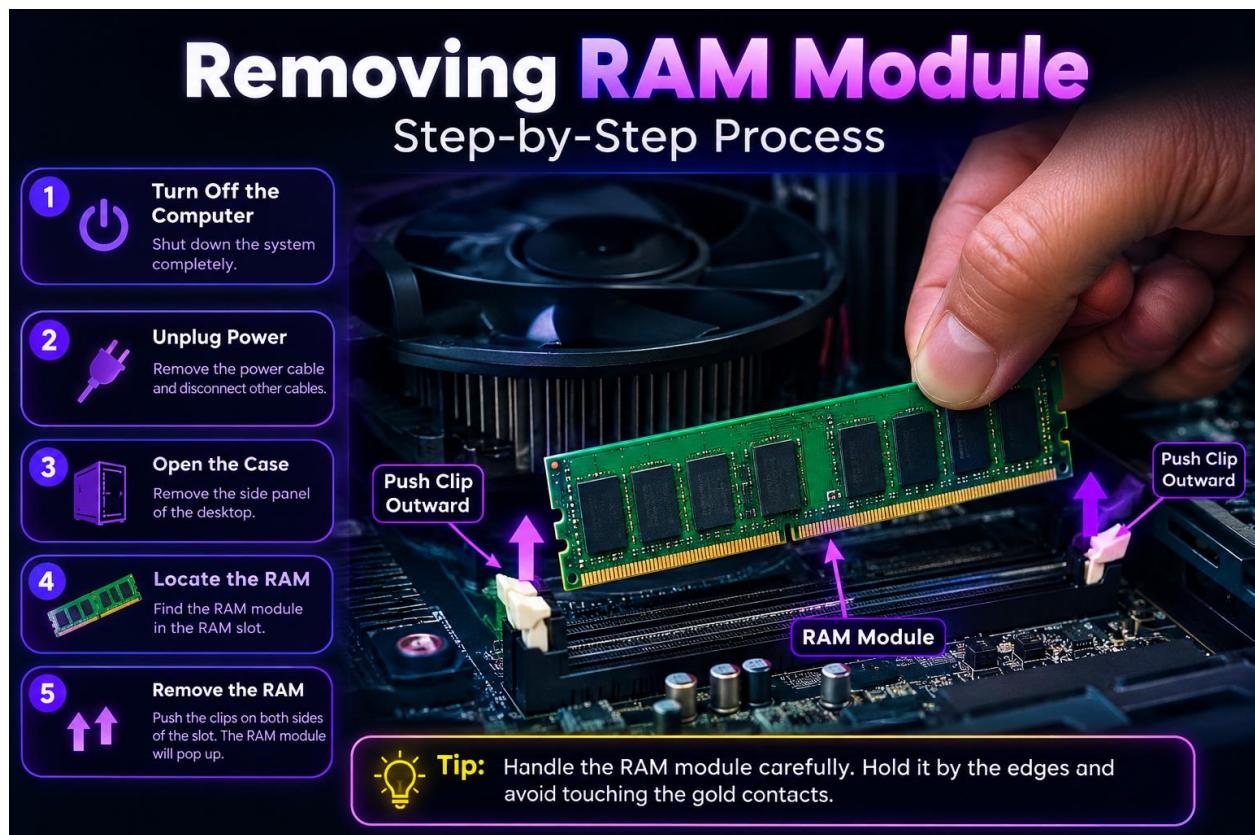


1 Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

❖ Ans ::



❖ **Shut down the computer**

Turn off the desktop/laptop completely.

❖ **Unplug the power cable**

Remove the power cable and disconnect other cables if necessary.

❖ **Open the computer case**

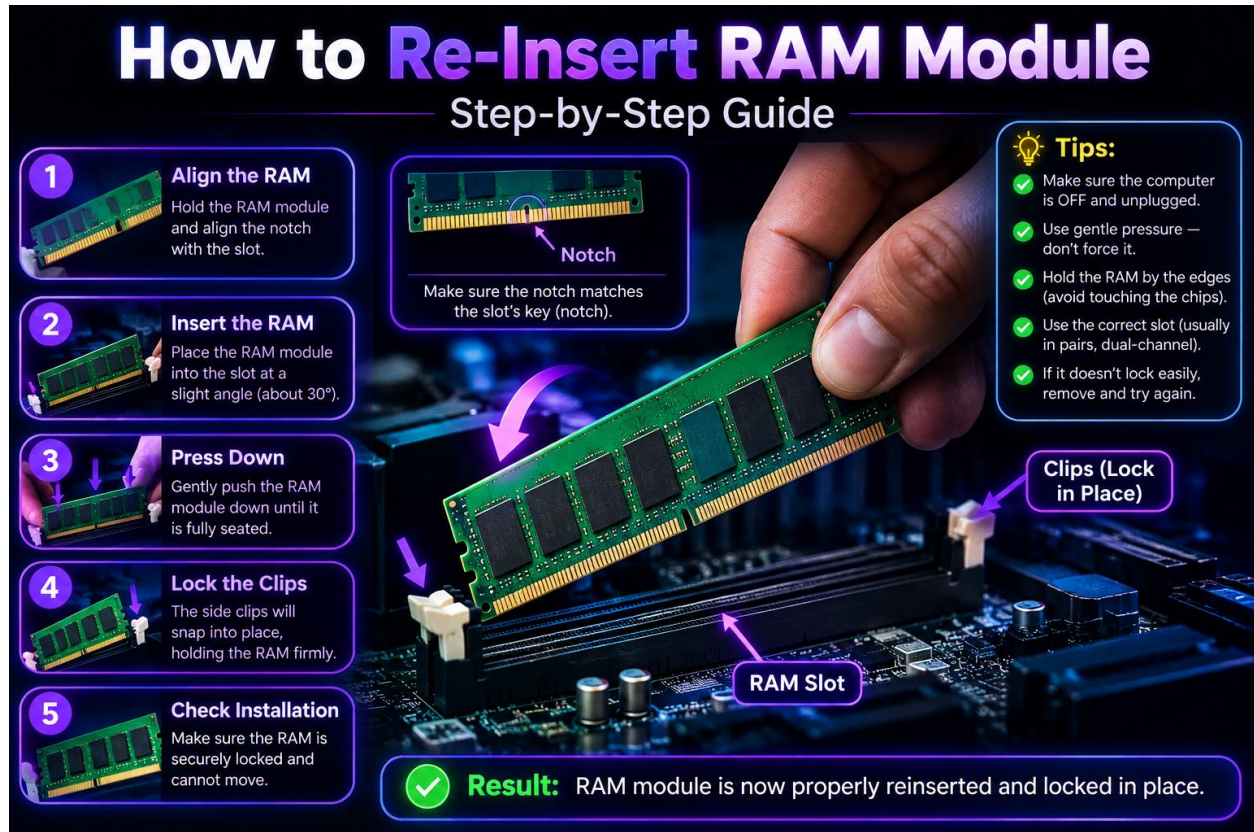
Remove the side panel of the desktop. For a laptop, remove the back cover if the RAM is accessible.

❖ Locate the RAM slots

Find the RAM module installed in the RAM slot on the motherboard.

❖ Remove the RAM

Open the clips on both sides of the RAM slot and carefully remove the RAM module.



❖ Check the RAM

Check the RAM for dust, damage, or any visible problem.

❖ Re-insert the RAM

Align the notch on the RAM with the slot and press it down firmly until the clips lock into place.

❖ **Close the computer case**

Put the side/back panel back on and reconnect all cables.

❖ **Turn on the computer**

Start the computer and check whether it boots normally.

Issues Observed

- No physical damage was observed on the RAM.
- There was some dust around the RAM slot.
- The RAM was reinserted correctly and locked properly.
- The computer started normally after reseating the RAM.

Result

The RAM reseating process was completed successfully, and the computer worked normally after restarting.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

GPU Troubleshooting Checklist

1. Identify the Problem

- Check for **no display**.
- Check for **screen flickering**.
- Check for **game crashes or low FPS**.

2. Check Display Cable

- Check the **HDMI/DisplayPort cable**.
- Make sure it is connected properly to the **GPU and monitor**.

3. Check GPU Power

- Check the **6-pin/8-pin power cable**.
- Make sure it is connected properly to the GPU.

4. Check GPU Installation

- Turn off the PC.
- Check that the GPU is **properly fitted in the PCIe slot**.

5. Check GPU Drivers

- Open **Device Manager**.
- Check the GPU driver.
- Update the driver if needed.

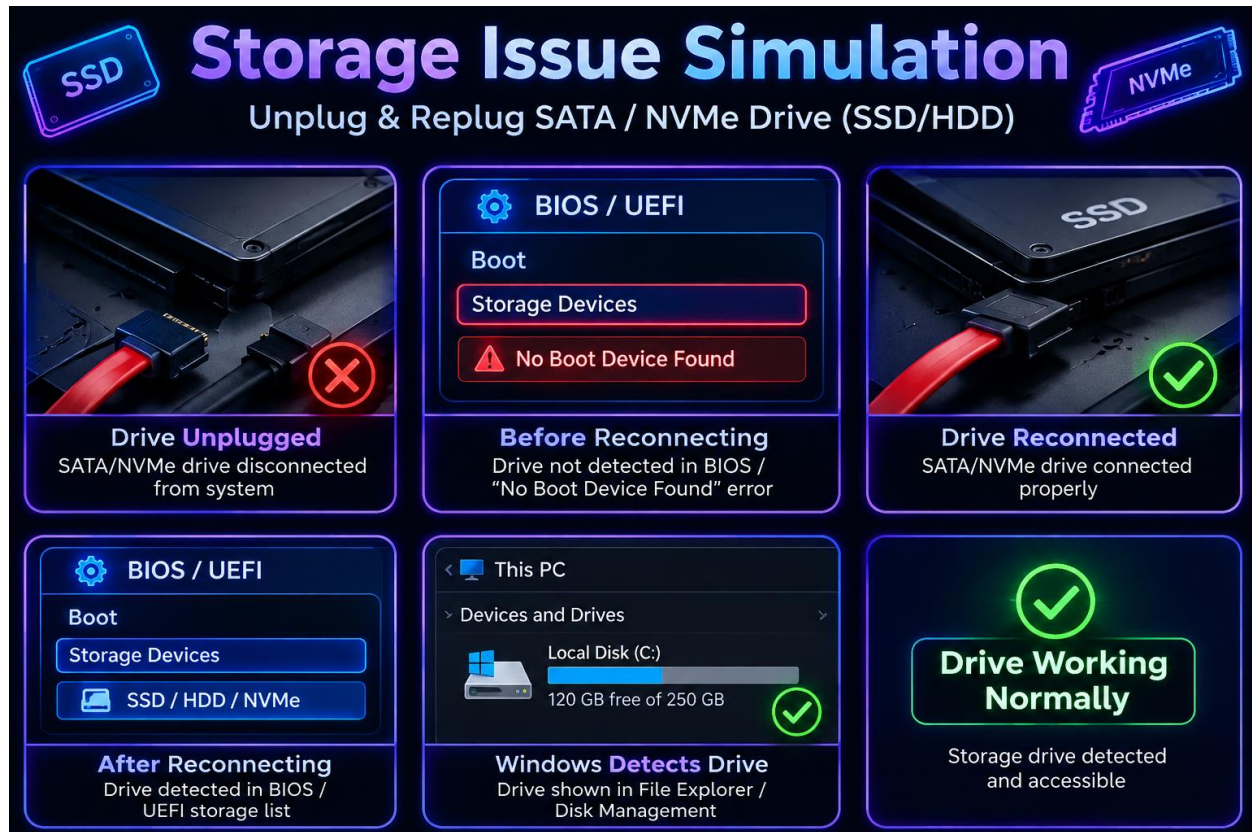
6. Check Temperature

- Check the GPU temperature while gaming.
- Make sure the GPU is **not overheating**.

7. Test the Fix

- Restart the PC.
- Start a game.
- Check that the **display and game work normally without crashes**.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.



Storage Issue Simulation

1. Shut down the PC

- Turn off the test computer completely.
- Unplug the power cable.

2. Disconnect the Drive

- For **SATA SSD/HDD**, disconnect the SATA data or power cable.
- For **NVMe SSD**, remove the NVMe drive carefully from its slot.

3. Start the PC and Check BIOS

- Turn on the PC and open **BIOS/UEFI**.
- The disconnected drive may **not appear in the Storage/Boot device list**.
- You may also see “**No Boot Device Found**” or “**Boot Device Not Found.**”

4. Reconnect the Drive

- Turn off the PC again.
- Reconnect the SATA cable/power cable or reinstall the NVMe drive properly.

5. Check BIOS Again

- Start the PC and open BIOS/UEFI.
- The SSD/HDD/NVMe should now **appear in the storage or boot-device list**.

6. Confirm the Fix

- Boot into Windows.
- Check **File Explorer / Disk Management** to confirm that the drive is detected and accessible.

Result

Before reconnecting: Drive was not detected / boot-device error appeared.

After reconnecting: Drive appeared in BIOS/UEFI and Windows detected it normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

PC Randomly Restarts

Which Component to Check First & Diagnostic Steps

Component to Check First



RAM

Reason

I would check the **RAM** first because faulty or loose RAM can cause the PC to randomly restart, freeze, or crash.



RAM

Diagnostic Steps for RAM

- 1** **Shut down the PC** and unplug the power cable.
- 2** **Remove the RAM** from the slot.
- 3** **Check the RAM and slot** for dust or damage.
- 4** **Re-insert the RAM** properly into the slot.
- 5** **If there are two RAM sticks,** test them one at a time.
- 6** **Run Windows Memory Diagnostic** to check for RAM errors.
- 7** **Restart and test the PC** to confirm that the random restarting problem is fixed.

1. Component to Check First: RAM

Reason:

I would check the **RAM first** because faulty or loose RAM can cause the PC to **randomly restart, freeze, or crash**.

2. Diagnostic Steps for RAM

1. **Shut down the PC** and unplug the power cable.
2. **Remove the RAM** from the slot.
3. **Check the RAM and slot** for dust or damage.
4. **Re-insert the RAM** properly into the slot.
5. If there are two RAM sticks, **test them one at a time**.
6. Run **Windows Memory Diagnostic** to check for RAM errors.
7. **Restart and test the PC** to confirm that the random restarting problem is fixed.